MILITARY SPECIFICATION

COLORS, TINTING, FOR INTERIOR NONFLAMING (DRY) PAINTS

1. SCOPE

1.1 This specification covers eight tinting colors (Rose Wood, Beach Sand, Yellow-Gray, Green-Gray, Pearl Gray, Sun Glow, Clipper Blue, and Pastel Blue) to be used by mixing with Paint, Non-Flaming (Dry), Chlorinated Alkyd Resin, Soft White, Semigloss, Formula No. 124/58 (Specification MIL-P-17970), to produce colored interior finish paints for living spaces on ships. Tinting colors covered by this specification shall be of but one type and of the color specified (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard for Testing Purposes.
TT-C-290 - Chrome-Yellow and Chrome-Orange, Dry (Paint-Pigments).
TT-D-643 - Drier, Paint, Naphthenate, Liquid, Concentrated.
TT-L-70 - Lampblack; Dry (Paint-Pigment).
TT-P-143 - Paint, Varnish, Lacquer and Related Materials; Packaging of.
TT-P-385 - Pigment, Iron-Blue, Dry.
TT-T-291 - Thinner, Paint, Volatile Mineral Spirits (Petroleum Spirits).
TT-T-562 - Toluidine-Red-Toner; Dry (Paint-Pigment).

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MIL-A-15206 - Aluminum Stearate.
MIL-P-17970 - Paint, Nonflaming (Dry), Chlorinated Alkyd Resin, Soft White, Semigloss, Formula No. 124/58.
MIL-R-21417 - Resin, Chlorinated Alkyd, Solution.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM-D1482-57T - Color Difference Using the General Electric Spectrophotometer.
ASTM-D1495-57T - Color Difference Using the Color Eye.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

FSC 8010

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3. REQUIREMENTS

3.1 Formula. -

3.1.1 The tinting colors shall consist of ingredients conforming to the applicable specifications in the proportions shown in table I, except that the pigment proportions may be varied slightly to conform to the color requirement. The exact formula used shall be shown on the forms required by the inspection procedure of method 1031 of Standard FED-STD-141 (see 4.2).

Table I - Tinting color formulas.

Ingredients	Pounds per 100 gallons [1]							
	Rose Wood	Beach Sand	Yellow Gray	Green Gray	Pearl Gray	Sun Glow	Clipper Blue	Pastel Blue
Chrome yellow (type II of Spec. TT-C-290)	320	282	202	122	33	625	150	---
Toluidine red (Spec. TT-T-562)	62	35	13.2	5.8	--	9	---	2.4
Lampblack (Spec. TT-L-70)	2.6	3.1	8.8	19.1	21	0.4	11.9	10.4
Iron blue (Spec. TT-P-385)	---	---	---	----	--	---	13.0	14.5
Magnesium silicate[2]	115	180	243	279	300	62	271	306
Aluminum stearate (Spec. MIL-A-15206)	10	10	10	10	10	10	10	10
Chlorinated alkyd resin (Spec. MIL-R-21417)	400	370	360	345	345	435	360	345
Petroleum spirits (grade 1 of Spec. TT-T-291)	236	262	273	286	295	210	273	295
Lead naphthenate drier (type I of Spec. TT-D-643)	5.0	4.7	4.5	4.4	4.3	5.5	4.5	4.3
Cobalt naphthenate drier (type II of Spec. TT-D-643)	2.1	1.9	1.9	1.8	1.8	2.2	1.9	1.8
Anti-skinning agent[3]	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

[1] The formula is given slightly in excess of 100 gallons to allow for normal manufacturing loss.
[2] The magnesium silicate shall be a micronized grade equal to "S.F. Magnesium Silicate" marketed by Whittaker-Clark and Daniels.
[3] Anti-skinning agent shall be equal to "Anti-oxidant B" marketed by National Aniline Division of Allied Chemical and Dye Corporation.

3.2 Manufacture. - The component raw materials shall be mixed and ground as required to produce a tinting color which is uniform, easily dispersed in the white base (Formula No. 124/58), free from grit, free from color separation, entirely suitable for the purpose intended, and in complete conformity to the requirements of this specification.

3.3 Quantitative requirements. - The tinting colors shall conform to table II when tested as specified in 4.3.2.

3.4 Qualitative requirements. -

3.4.1 Odor. - The odor shall be normal for the volatiles permitted.

3.4.2 Color. - When the contents of a one-half pint can of a tinting color is mixed as directed in 4.3.2.2 with one gallon of Soft White Chlorinated Alkyd Resin Paint (Formula No. 124/58), the color of the dry film of the resultant paint shall acceptably match the applicable standard color card (see 6.3).

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Table II - Quantitative requirements.

	Percent by weight	
	Minimum	Maximum
Water	--	0.5
Coarse particles and skins (as residue retained on standard No. 325 sieve) (Spec. RR-S-366)	--	0.5
Viscosity, Krebs units	67	77
Fineness of grind	6	----

3.4.3 Compatibility. - There shall be no evidence of incompatibility of any of the ingredients of the tinting color or of the resultant paint when mixed as described in 4.3.2.2.

3.4.4 Condition in container. - For at least one year after date of manufacture, the tinting color shall be capable of being readily broken up with a paddle to a smooth, uniform consistency; shall not liver; shall not exceed 90 Krebs units in viscosity; shall not curdle, gel, nor show any other objectionable properties.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling, inspection and testing. - Sampling, inspection, and testing shall be in accordance with method 1031 of Standard FED-STD-141 as hereinafter supplemented. Submission of forms in accordance with 2.2.1.3.5 and 2.2.1.3.5.11 of method 1031 shall cover complete tests other than the one year condition in container test on the first lot of material submitted under each contract. Also, this lot shall be one of those selected for complete tests with the above exception, in accordance with 2.2.1.2.5 of method 1031. The condition in container requirement shall apply for one year after manufacture regard less of other testing or prior acceptance of the material. Inspection shall particularly determine precision of fill.

4.2.1 Ingredient materials. - One pint of the chlorinated alkyd resin, and when requested by the Laboratory Superintendent or other authority, samples of the following ingredient materials of at least the amount shown shall be taken by the inspector for test purposes:

1/2 pint chrome yellow.
1/2 pint toluidine red.
1/2 pint lampblack.
1/2 pint iron blue.
1/2 pint magnesium silicate.
1/2 pint aluminum stearate.
1 pint petroleum spirits.
1/4 pint lead naphthenate drier.
1/4 pint cobalt naphthenate drier.
1/4 pint anti-skinning agent.

4.3 Test procedures. -

4.3.1 Ingredient materials. - All ingredient materials submitted shall be tested to determine compliance with the applicable specifications listed in table I.

4.3.2 Finished tinting color. -

4.3.2.1 The following tests shall be conducted in accordance with the methods specified in Standard FED-STD-141:

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Tests	Method
Water	4081
Coarse particles and skins	4092
Viscosity	4281
Fineness of grind	4411
Odor	4401

4.3.2.2 Color. - Add the entire contents of the one-half pint can of tinting color to one gallon of Soft White Chlorinated Alkyd Resin Paint (Formula No. 124/58) and mix thoroughly. (A suggested procedure is as follows: After draining the one-half pint can of tinting color into the soft white paint with constant stirring, partly refill the one-half pint can with some of the mixed paint, stir, and return to the gallon of mixed paint. Stir until completely uniform.) Prepare a dry film of the mixed paint by using a film applicator (doctor blade) to draw down a film on a plane piece of opaque white glass at a wet film thickness of approximately 0.003 inch. After a 48-hour air drying period, compare the color of the film with the pertinent standard color card (see 6.3) as specified in method 4250 of Standard FED-STD-141. If doubt exists as to the satisfactoriness of the match, the color shall show no more than two color units difference in the direction of minus ΔL, Δa and Δb values from the standard when tested in accordance with ASTM method D1482-57T, ASTM method D1495-57T or similar procedure using an instrument of equal sensitivity.

4.4 Additional inspection. - Where other specifications form a part of this specification, unless otherwise specified in the contract or order, sampling, examination, and tests shall be conducted as required in the referenced specification.

4.5 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - The tinting colors shall be packaged level A or C; packed level A, B or C and marked as specified (see 6.1) in accordance with Specification TT-P-143. The tinting colors shall be furnished in the quantity specified (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color of tinting color required (see 1.1).
(c) Size of container.
(d) Levels of packaging and packing required (see 5.1).

6.2 Tinting colors should be purchased by volume, the unit being a U.S. one-half pint at 15.6°C. (60°F.).

6.3 Color cards. - Color cards for the mixed paint (tinting color plus soft white, Formula No. 124/58) may be obtained upon application to the Chemical Laboratory, Norfolk Naval Shipyard, Portsmouth, Va. When requesting specify for what purpose cards are desired.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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